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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	JONG JAVA
Project / Solution Name	JOBRIDGE
Name (Leader)	Louis Benedict Oen
Email & Phone (Leader)	louisbenedict127@gmail.com (+60 11 2371 3971)
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://jobridge.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/louisbenedictt/Jobridge

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

JOBRIDGE has two parts. First, an offline data pipeline (Jobridge_pipeline.ipynb, Python in Jupyter) downloads every monthly Hacker News "Ask HN: Who is hiring?" thread (January 2020 to October 2026, 40,067 job posts) from the public HN Algolia API and reads the Kaggle LinkedIn Job Postings dataset (123,849 postings). It detects 19 job roles and 128 skills with a curated regular-expression dictionary, computes an AI Adoption Index, forecasts each skill's share of job posts under a moderate and a fast AI scenario (scikit-learn models selected by rolling backtest), builds role skill profiles, maps roles to ILO generative-AI exposure scores through ISCO-08 codes, and links roles and skills to UN SDG targets. Results are saved as CSV/JSON files. Second, the web application: a browser front end (HTML) calls a Flask JSON API (flask_app.py), which uses core.py to extract skills from an uploaded or pasted CV, score role matches, build a reskilling pathway from a checked course catalogue, re-rank roles by the user's SDG priorities and produce the HR team view. There is no database and no AI API call at run time. The app is developed and hosted on Replit, code is kept on GitHub.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12.4	Language of the data pipeline and the backend	PSF License

Name & Version	Purpose in Project	License
JavaScript, HTML5, CSS3	Front end (static/jobridge.js, templates/jobridge.html)	Open web standards
Jupyter (ipykernel)	Running the data pipeline notebook	BSD-3-Clause
Flask 3.1.3 (with Jinja2 3.1.6, MarkupSafe 3.0.3)	Web server, page template and JSON API	BSD-3-Clause
pandas 3.0.1	Data tables, monthly time series, CSV processing	BSD-3-Clause
NumPy 2.4.3	Numerical Computation	BSD-3-Clause
scikit-learn 1.8.0	Forecasting models (HistGradientBoostingRegressor, Ridge) and scaling	BSD-3-Clause
pypdf 6.21.1	Reading text from uploaded PDF CVs	BSD-3-Clause
requests 2.32.5	Downloading Hacker News threads from the HN Algolia API (pipeline only)	Apache-2.0
matplotlib 3.10.8	Charts inside the pipeline notebook	Matplotlib License (PSF-based)
Plotly 7.1.0 (Python) with bundled plotly.js	Interactive charts in the web app	MIT
Tailwind CSS 3 (Play CDN)	Styling of the user interface	MIT
Font Awesome Free 6.4.0 (via cdnjs)	Icons	Icons CC BY 4.0, fonts SIL OFL 1.1, code MIT
Google Fonts: Inter, Plus Jakarta Sans, IBM Plex Mono	Typography	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
HN Search API (hn.algolia.com/api), Algolia for Y Combinator Hacker News	Download of the monthly "Who is hiring?" job posts by the data pipeline (offline; not called by the running app)	Free, public, no key
Google Fonts (Google)	Web fonts loaded by the user's browser	Free
cdnjs (Cloudflare) and Tailwind Play CDN	Font Awesome icons and Tailwind CSS loaded by the user's browser	Free
Generative-AI / LLM APIs	None. The application does not call any GenAI or LLM API.	None

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development workspace and hosting of the live prototype (Replit Deployment running flask_app.py)	Replit Core (SDC sponsor-provided)
GitHub	Code repository and version control	Free
Kaggle	Source of the LinkedIn Job Postings dataset (download only)	Free account

Service & Provider	Purpose in Project	Plan / Region
Database / storage / authentication services	None. Data are files in the repository; no user accounts.	None

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gradient boosting regressor (scikit-learn HistGradientBoosting Regressor, 200 iterations): forecasts each skill's share of job posts 6 and 12 months ahead under two AI scenarios; selected by rolling backtest; used for data cutoffs from 2023-06	Team, built with scikit-learn 1.8.0	Self-trained on monthly Hacker News skill shares	BSD-3-Clause (library); model trained by the team
"Trend plus AI" linear model: skill trend plus AI Adoption Index term; used for the earlier cutoffs in the time-machine view	Team	Self-trained	Team-owned
Ridge regression, linear trend and naive forecast: baselines compared in the backtest, not used for the app's forecasts	Team, built with scikit-learn / NumPy	Self-trained	BSD-3-Clause (library)

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Skill and role detection: 128 skill and 19 role regular-expression patterns (rule based, not a machine-learning model)	Team	Hand-curated rules	Team-owned
Pre-trained / generative AI models (LLMs etc.)	None embedded in or called by the application.		

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Hacker News "Ask HN: Who is hiring?" monthly threads, 2020-01 to 2026-10 (40,067 job posts)	https://hn.algolia.com/api	Public	Hacker News content via public API; raw text not redistributed, only aggregates stored	N, in stored data (raw posts can contain recruiters' contact e-mails; these are not kept)

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
LinkedIn Job Postings 2023–2024, file postings.csv (123,849 postings listed 24 Mar–20 Apr 2024)	https://www.kaggle.com/datasets/arshkon/linkedin-job-postings	Public	Kaggle dataset licence	N
ILO Working Paper 140 (2025), Generative AI and Jobs: occupational exposure index, Table A1 (427 occupations)	https://webapps.ilo.org/static/english/intserv/working-papers/wp140/index.html	Public	© ILO 2025, cited	N
ISCO-08 occupation classification	https://www.ilo.org/public/english/bureau/stat/isco/isco08/index.htm	Public	ILO reference, cited	N
UN Sustainable Development Goals and targets	https://sdgs.un.org/goals	Public	United Nations, cited	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Course catalogue: 127 free or low-cost courses (Alison, OpenLearn, Microsoft Learn, Kaggle, Cisco Networking Academy, Atlassian, HRD Corp e-LATiH and others), checked 22–24 Oct 2026	Provider pages (data/reference/courses.csv)	Self-collected	Links and metadata only; course content belongs to each provider	N
Skill taxonomy, role patterns, AI terms and SDG links (128 skills, 19 roles, 100 SDG links), informed by ESCO and O*NET skill groupings	data/reference	Self-created (AI-assisted, see 4.3)	Team-owned	N
Synthetic sample CVs (8)	data/cvs	Synthetic	Team-owned	N (fictional people)

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
TalentCorp Malaysia, Impact Study of AI, Digital and Green Economy on the Malaysian Workforce	https://www.mymahir.my	Public	Background context only; no figures used in scoring	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude (Anthropic), chat assistant	Drafting the data pipeline code, skill dictionary, role patterns, course search, SDG links and synthetic sample CVs (recorded in data/reference/sources.csv and sdg_links.csv)	Major
Claude Code (Anthropic), model Claude Opus 5.5	Connecting the JOBRIDGE front end to the Flask backend (JSON API in flask_app.py, static/jobridge.js), automated tests, debugging, user-interface changes and deployment notes	Major
Replit Agent	Deployment	Major
Gemini	Help design frontend	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team ran the pipeline and the app end to end and checked the outputs against the source files. 19 automated tests confirm that every number shown in the app equals the calculation in core.py, and a fairness test confirms that two CVs differing only in name, gender and age receive identical recommendations. Each course link was opened and checked by hand.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core (US$20 per participant)	Development workspace, Replit Agent, and hosting of the live prototype through Replit Deployment
RunPod	US$25 team credit	Not used: JOBRIDGE has no GPU workload.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is where the team developed and deployed the live prototype, as the competition requires. JOBRIDGE itself is a standard Python/Flask application whose data are plain CSV/JSON files, so it does not depend on any Replit-specific service and could run on any Python host. No sponsor GenAI API or GPU is needed for its core functionality.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Users may upload (.txt or .pdf) or paste a CV. The server reads it in memory to recognise skills; the CV is not stored, written to disk, or sent to any third-party or AI service. There is no database and there are no user accounts. Before matching, lines with personal details (name, gender, sex, age, date of birth, nationality, religion, race, marital status) and the interest section are removed, so they never influence any score. Only the list of recognised skills is kept, in the user's open browser page, and it disappears when the page is closed. HR team uploads are handled the same way. The browser saves one preference (sidebar open or closed) in localStorage. Fonts, icons and CSS are loaded from Google Fonts, cdnjs and the Tailwind CDN, which see the visitor's IP address. Raw Hacker News and LinkedIn texts are not redistributed; only aggregates are stored.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Skill trends come from Hacker News, which leans towards US technology companies, so non-tech and Malaysian demand are under-represented; the LinkedIn data (role profiles only) covers one month (Mar–Apr 2024). Skill detection uses a fixed regular-expression dictionary: it never invents skills but can miss synonyms or misread phrases (e.g. "SQL Accounts"). The forecasting models do not beat a naive forecast on level error, so forecasts are shown as direction and moderate-to-fast scenario ranges, not exact values. ILO scores measure exposure to generative AI, not job loss, and job titles are mapped to ISCO codes by the team. Mitigations: limited-data warnings for roles with fewer than 50 job posts, a time-machine view, a fairness test, a cited source for every number, and the final decision left to the user. No language model runs in the app, so there is no hallucination risk at run time.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

✔	All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
✔	All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
✔	No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
✔	Any use of generative-AI tools for development has been disclosed and reviewed by the team.
✔	Any use of official SDC sponsor resources has been declared in Section 5.
✔	The prototype described here is accessible to judges through the links provided in Section 1
✔	All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[24-10-2026]